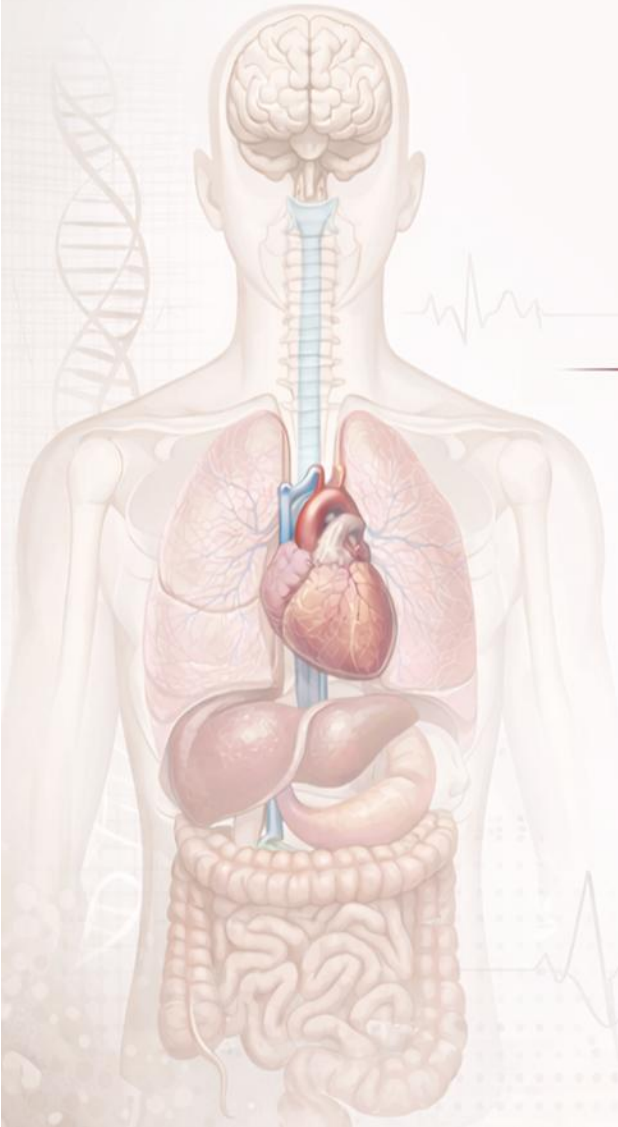
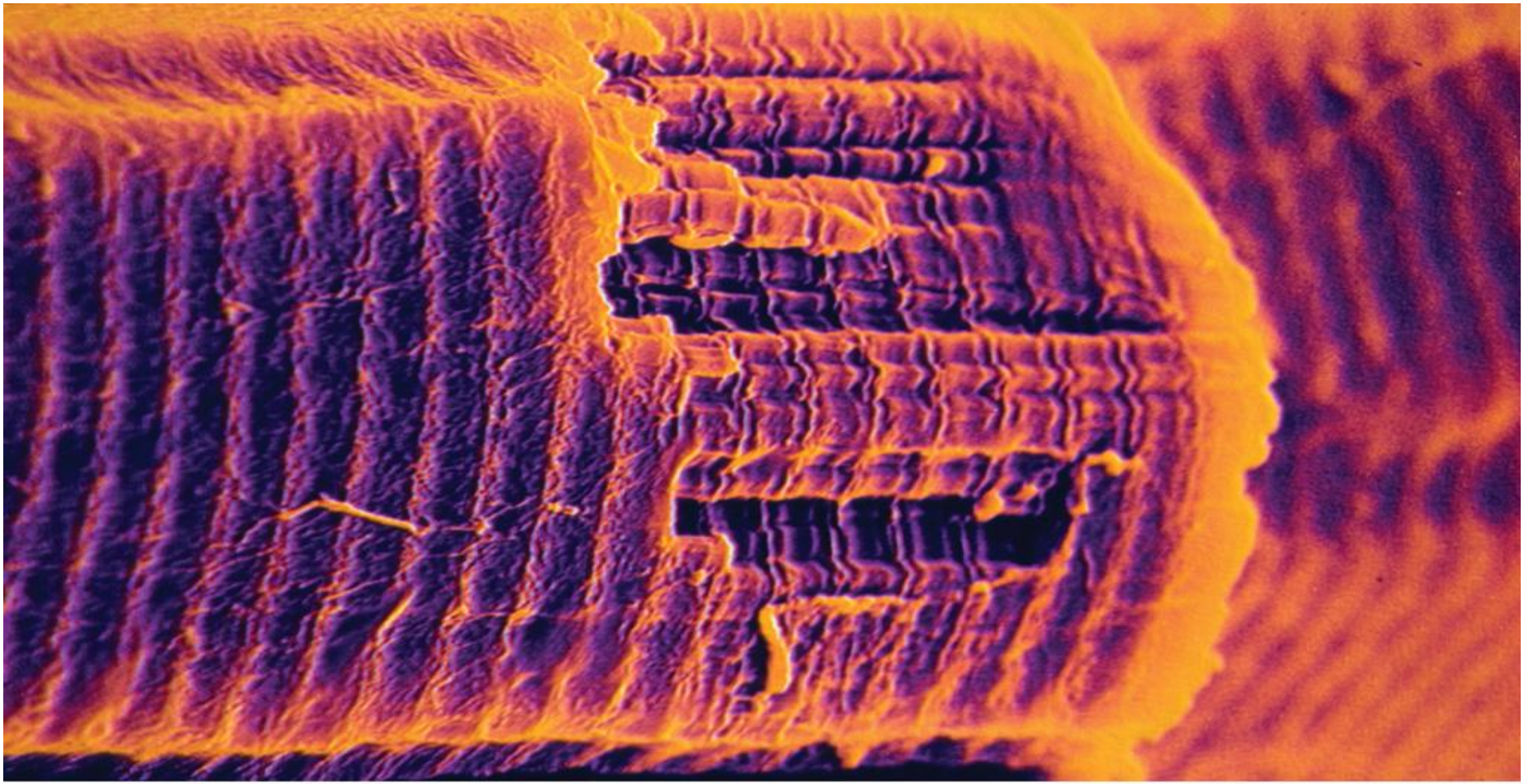


Chapter 8.
Muscle Physiology

KOREA UNIVERSITY
1905

A scanning electron micrograph of skeletal muscle fiber



© CNRI/Science Source

Muscles in terms of homeostasis

- Muscles are contraction specialists
- Three types of muscles
  - Skeletal muscles: attaches to the skeleton
    - Their contraction moves bones, allowing the body to perform diverse motor activities
    - Energy: acquiring, chewing, swallowing foods, breathing
    - Body temperature: heat-generating
    - Move away from harm, nonhomeostatic activities (dancing, work)
  - Smooth muscles: in the walls of hollow organs and tubes
    - Regulates movement of blood (blood vessels), food (digestive tract), air (respiratory airway), urine (urinary system)
    - Eye
  - Cardiac muscles: only in the walls of the heart
    - Contraction pumps blood throughout the body

8.1 Structure of Skeletal Muscle

Muscle tissues and sliding filament model

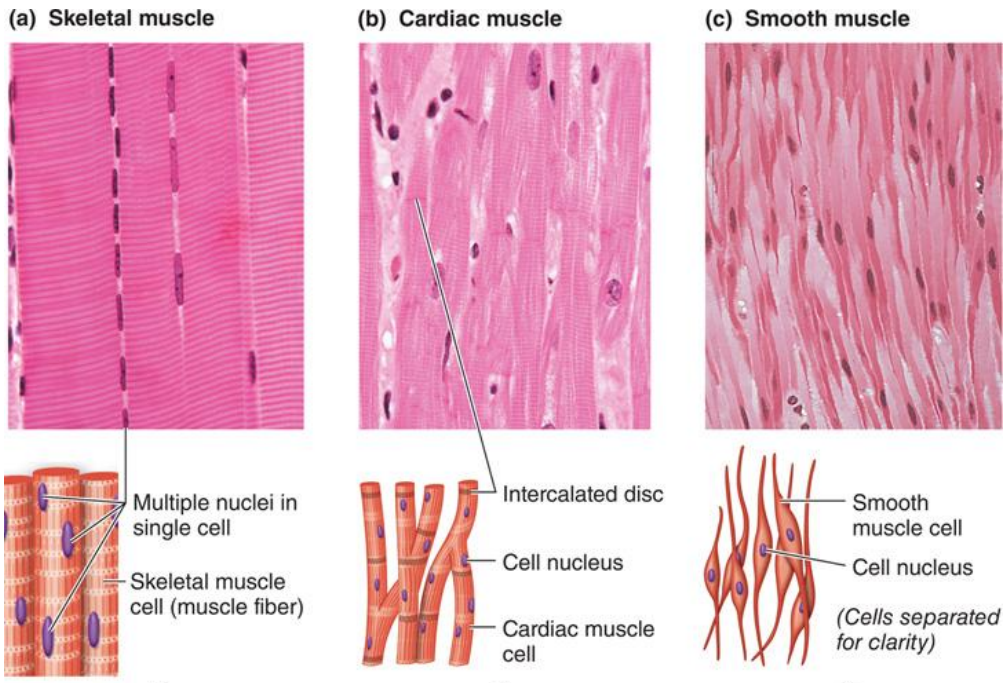


https://www.youtube.com/watch?v=dpxalWACO7k

Structure of skeletal muscle

- By moving specialized intracellular components, controlled muscle contractions allow
  - Purposeful movement of the whole body or parts of the body
  - Manipulation of external objects
  - Propulsion of contents through various hollow internal organs
  - Emptying of organ contents to the external environment
- Muscle comprises largest group of tissues in the body, 50% w.
- Three types of muscles can be classified in two ways
  - Structure) Striated (skeletal & cardiac) or unstriated (smooth)
    - Alternating dark and light bands or stripes are found (skeletal, cardiac) or not (smooth)?
  - Control) Voluntary (skeletal) or involuntary (cardiac, smooth)
    - Innervated by somatic (skeletal) or autonomic nervous system (cardiac, smooth)
    - Although skeletal muscles are categorized as voluntary, much skeletal muscle activity is also under subconscious involuntary control like posture, balance, rhythmic movement, and reflex

Characteristics of three types of muscle



Classification: Striated muscle, voluntary muscle

Description: Bundles of long, thick, cylindrical, striated, contractile, multinucleate cells that extend the length of the muscle

Typical location: Attached to bones of skeleton

Function: Movement of body in relation to external environment

Classification: Striated muscle, involuntary muscle

Description: Interlinked network of short, slender, cylindrical, striated, branched, contractile cells connected cell to cell by intercalated discs

Location: Wall of heart

Function: Pumping of blood out of heart

Classification: Unstriated muscle, involuntary muscle

Description: Loose network of short, slender, spindle-shaped, unstriated, contractile cells that are arranged in sheets

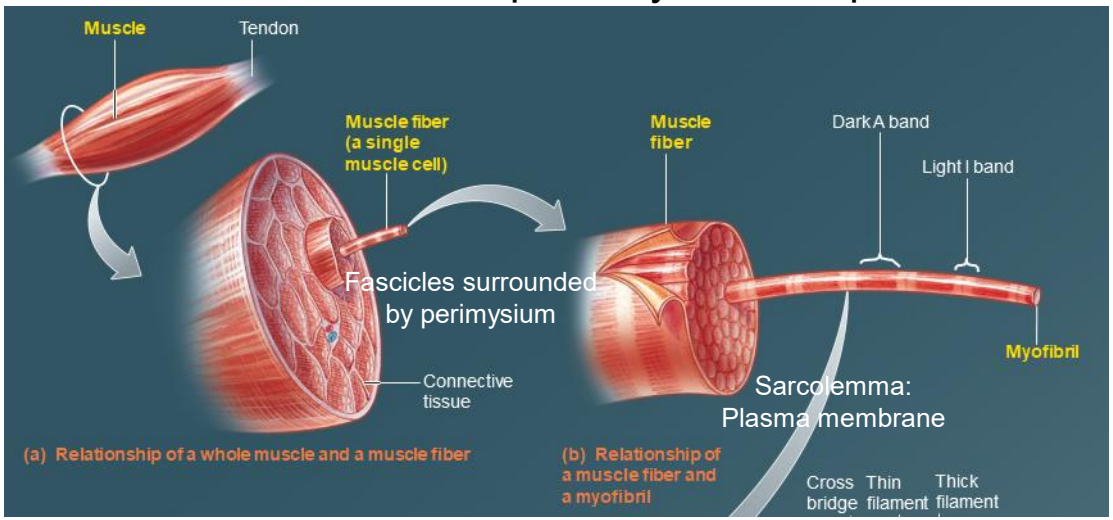
Typical location: Walls of hollow organs and tubes, such as stomach and blood vessels

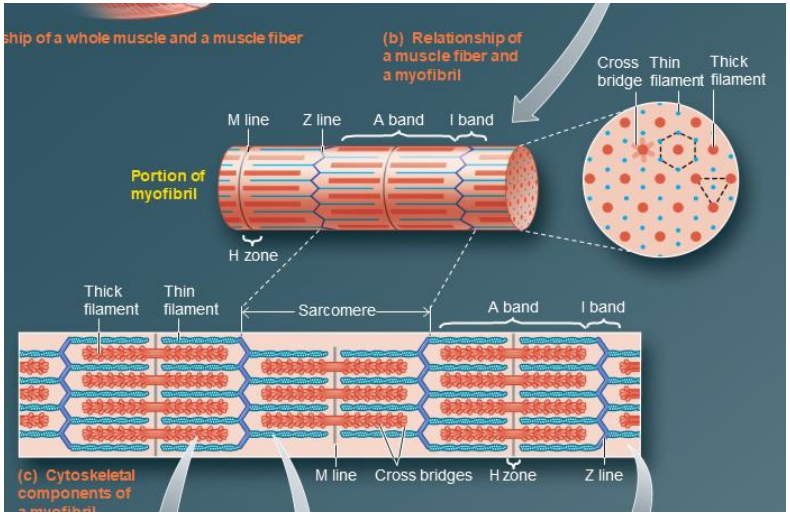
Function: Movement of contents within hollow organs

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Skeletal muscle fibers

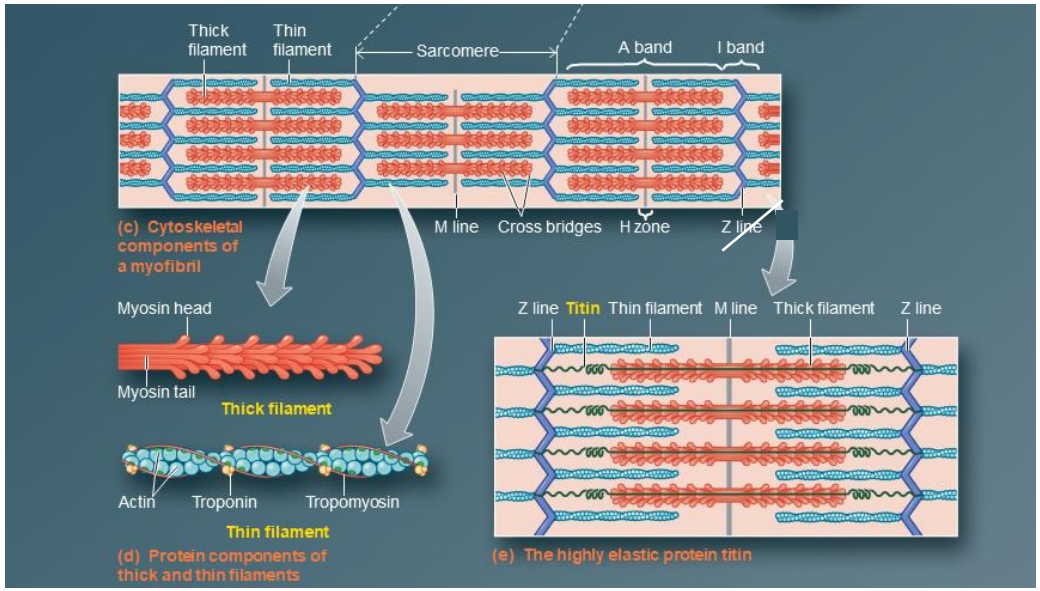
- **Muscle fiber**: a single skeletal muscle cell
  - Multinucleated cell with many mitochondria
- Structure of skeletal muscle: a number of muscle fibers lying parallel to one another and bundled together by connective tissue (perimysium) -> fascicle
- Fascicle: bundle of muscle fibers surrounded by perimysium
- **Myofibril**: regular arrangement of thick and thin microfilaments
  - Thick filament: special assembly of protein **myosin**
  - Thin filament: primarily made of protein **actin**

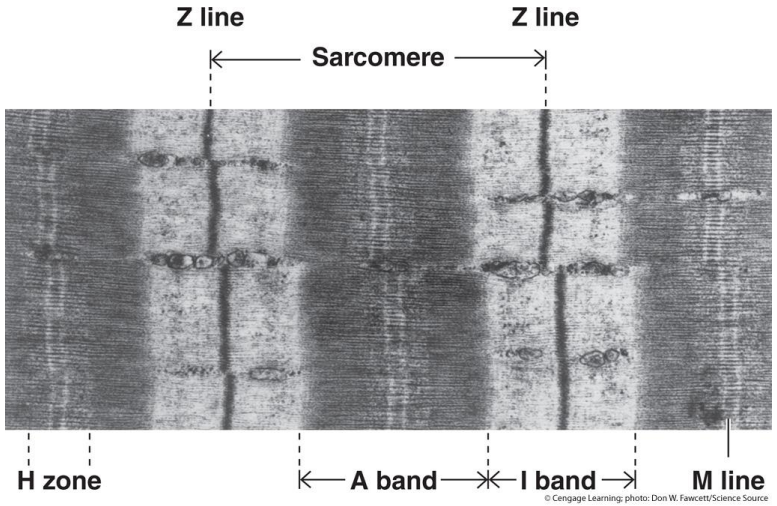




Bands and cross bridges

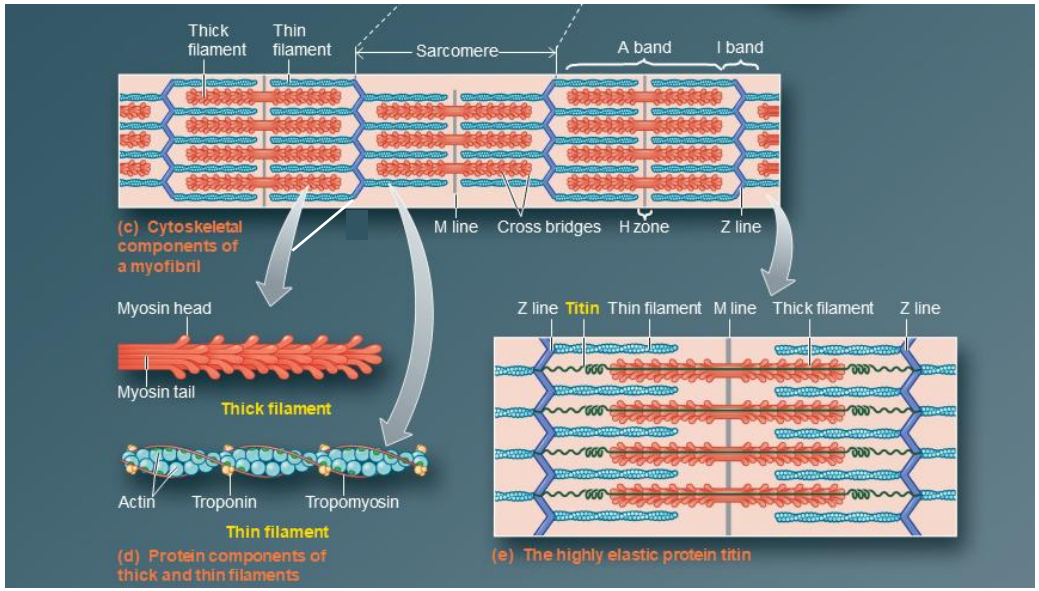
- **A and I bands**: alternating dark (A) and light (I bands) under electron microscope
  - A band: thick filaments together with portions of thin filaments that overlap
  - I band: remaining portion of thin filaments do not project into A band
  - Striations arise from the repeating pattern of A and I bands in aligned **sarcomeres**
  - H zone: middle of the A band where thin filaments do not reach
  - M line: central line in H zone composed of structural supporting proteins
  - Z line: dense vertical line visible in the middle of I band
  - Sarcomere: the area between two Z lines, **functional unit** of skeletal muscle





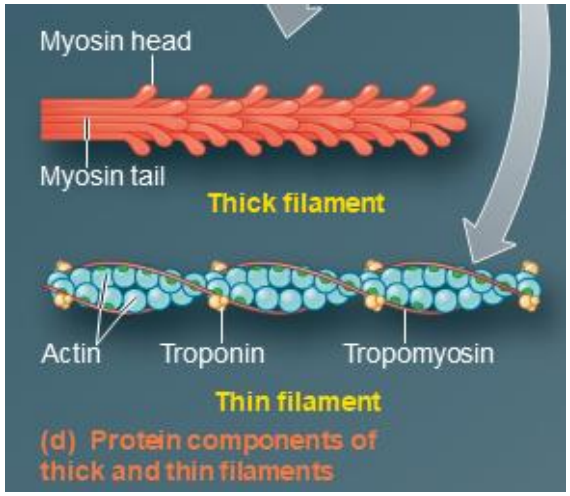
Bands and cross bridges

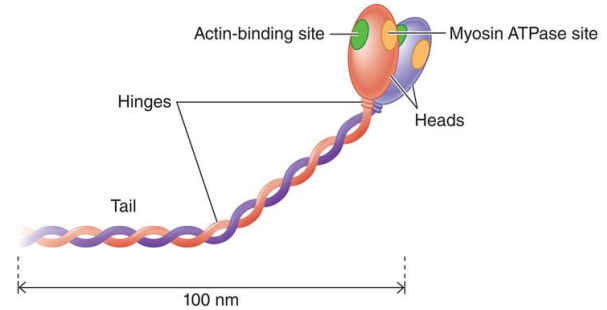
- **Titin**: single strand of giant, highly elastic protein connecting Z line, M line, and Z line
  - Serving as scaffold to stabilize the position of thick filaments in sarcomere
  - Acting as an elastic spring to increase muscle's elasticity
  - Participating in signal transduction (muscle enlargement in response to weight-lifting)
- **Cross bridges**: projection from each thick filament in six directions toward the surrounding thin filaments



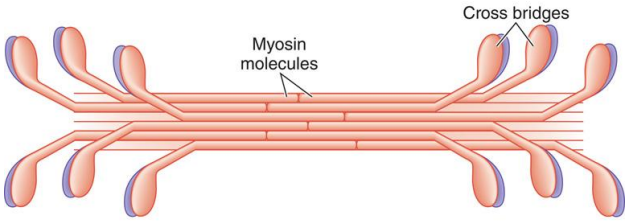
Myosin in the thick filament

- **Myosin**
  - Each thick filament has hundreds of myosin molecules in a specific structure
  - Myosin consists of two identical subunits, each in shape of golf club
  - The tails are intertwined around each other, with globular heads projecting outward
  - Two hinges in myosin molecule: one in the “neck” and one in the middle of tail
  - The heads form the cross bridges between the thick and thin filaments
    - Actin-binding site: binding to actin in thin filament
    - Myosin ATPase site: ATP-splitting site





(a) Myosin molecule



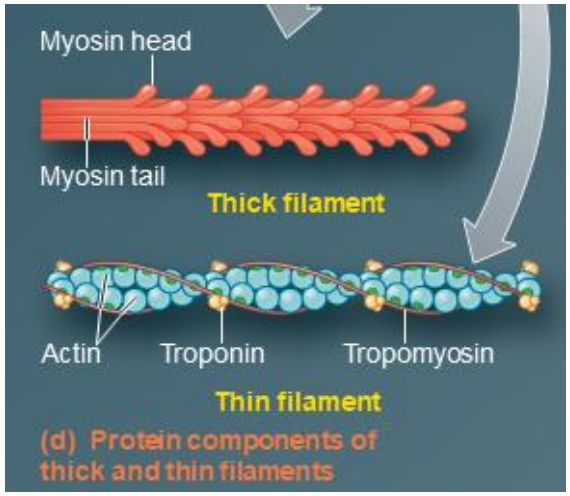
(b) Thick filament

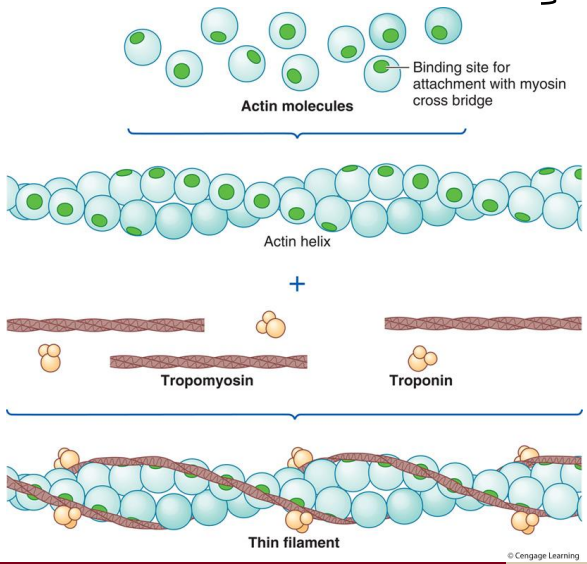
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Actin in the thin filament

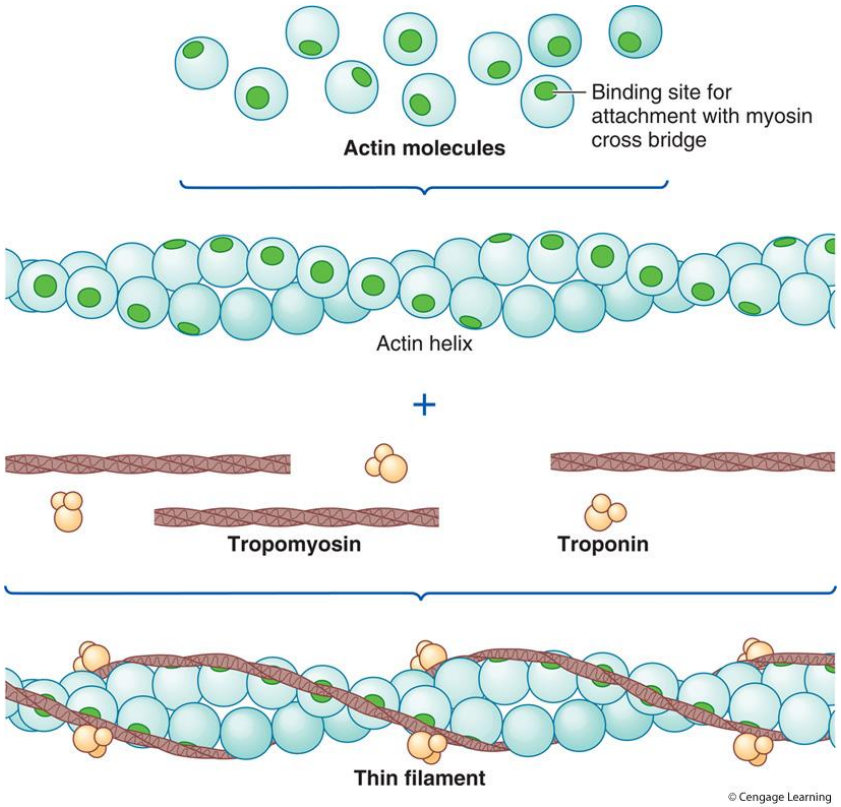
- **Actin**
  - Main structural component of the thin filament (+ tropomyosin + troponin)
  - Thin filaments backbone: spherical actin molecules joined into two strands and twisted (intertwined strings of pearls)
  - Binding site for myosin cross bridges
  - Binding of actin and myosin cross bridges -> contraction of muscle fiber
  - In relaxed muscle fiber, contraction does not take place in aid of..
  - Tropomyosin: covers the actin sites that bind with the cross bridges
  - Troponin: three portions bind to tropomyosin, actin, and Ca^{2+}

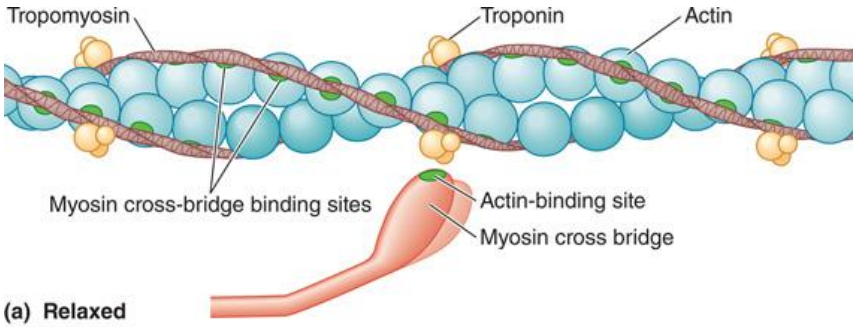
Regulatory proteins for muscle contraction





Actin in the thin filament



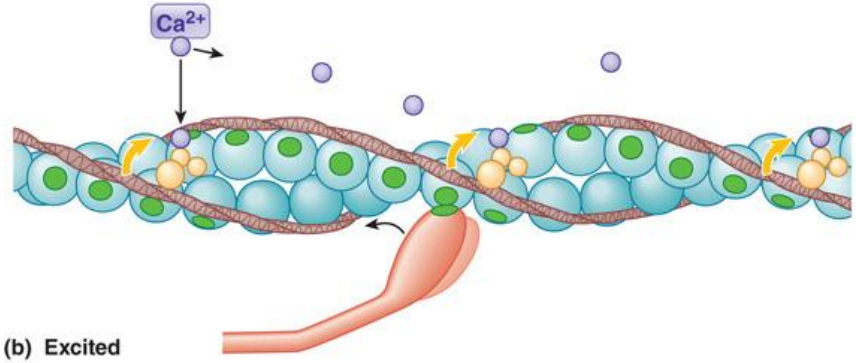


(a) Relaxed

1 No excitation.

2 No cross-bridge binding because cross-bridge binding site on actin is physically covered by troponin–tropomyosin complex.

3 Muscle fiber is relaxed.



(b) Excited

1 Muscle fiber is excited and Ca^{2+} is released.

2 Released Ca^{2+} binds with troponin, pulling troponin–tropomyosin complex aside to expose cross-bridge binding site.

3 Cross-bridge binding occurs.

4 Binding of actin and myosin cross bridge triggers power stroke that pulls thin filament inward during contraction.

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8.2 Molecular Basis of Skeletal Muscle Contraction

Cross-bridge cycling



https://www.youtube.com/watch?v=BVcgO4p88AA

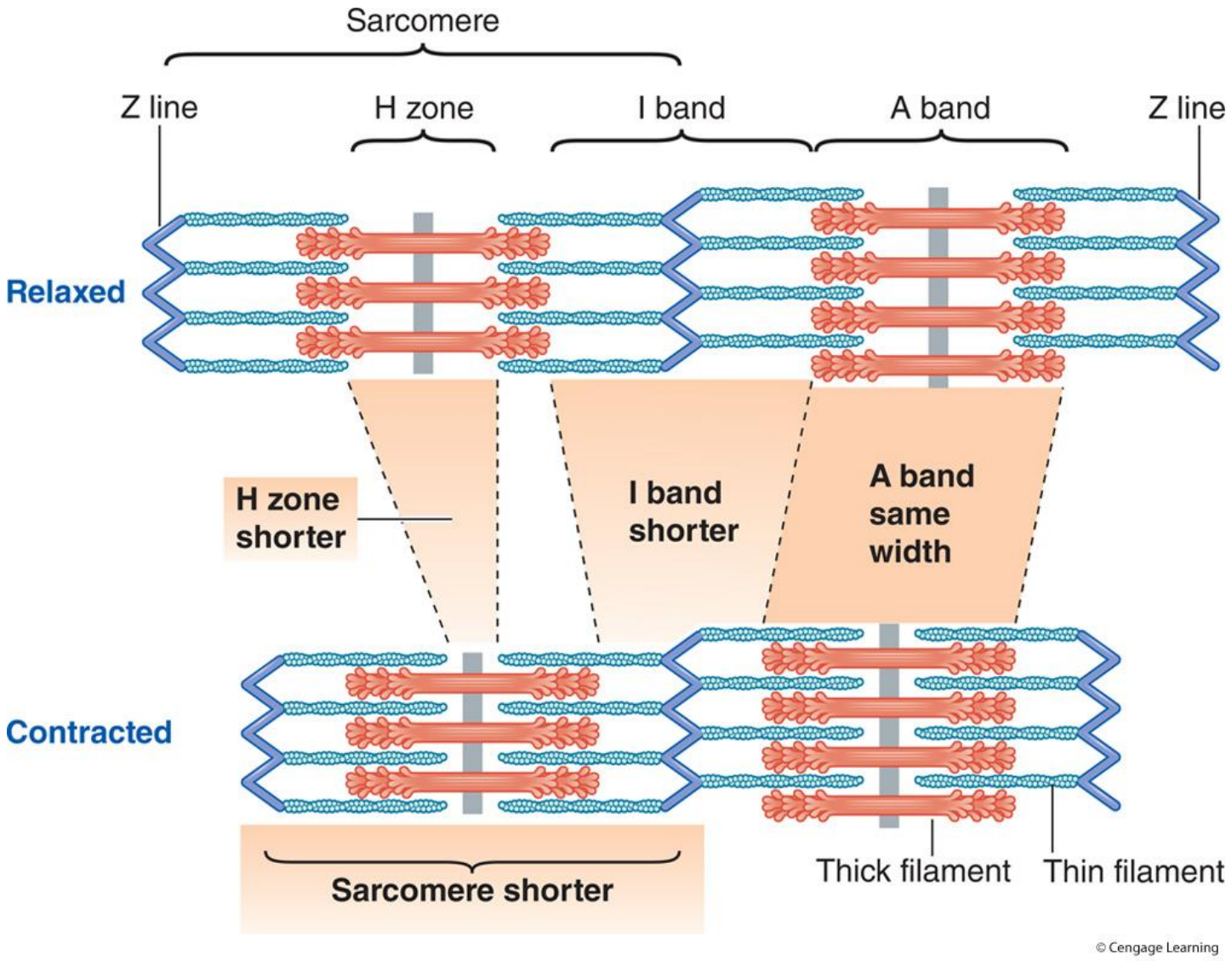
Cycles of cross-bridge binding and bending

In the molecular biology aspect,

- How does cross-bridge interaction between actin and myosin bring about muscle contraction?
- How does action potential trigger this contraction process?
- Source of Ca^{2+} repositioning troponin and tropomyosin?

- **Sliding filament mechanism**
- Muscle contraction is achieved by thin filaments from the opposite sides of each sarcomere sliding closer between the thick filaments, toward the A band's center
- Their sliding pulls the Z lines closer together, so the sarcomere shortens
- Width of H zone and I band gets shorter, while that of A band remains the same
- Muscle contraction is enabled by each sarcomeres' sliding closer together, not any of thick or thin filaments' length decreasing

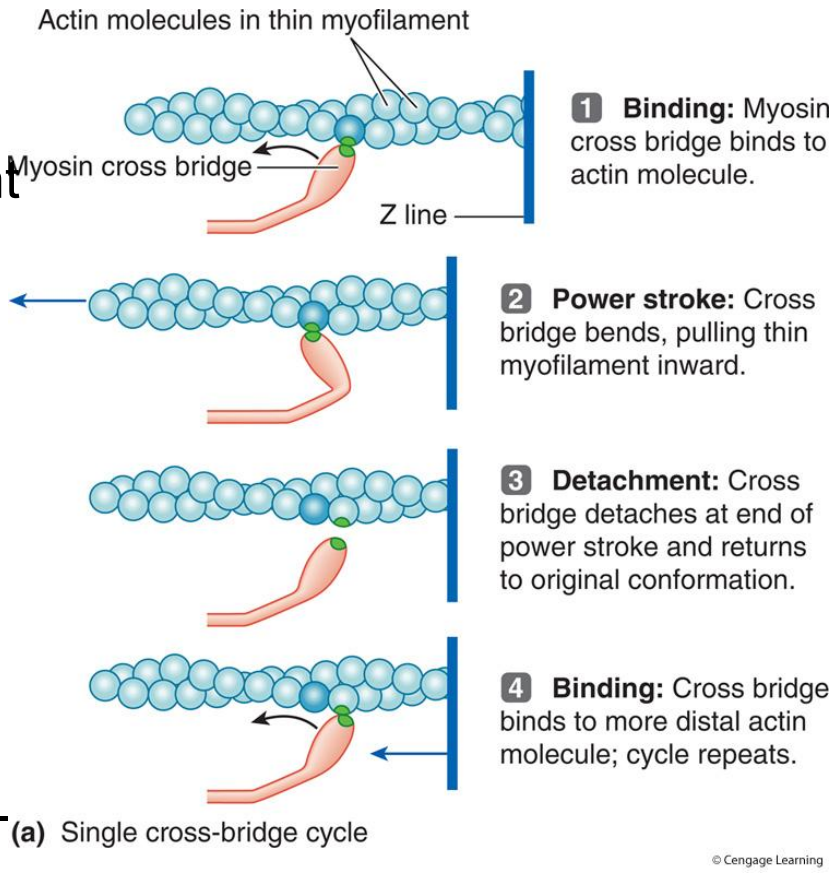
Cycles of cross-bridge binding and bending



Power stroke (at the single myosin level)

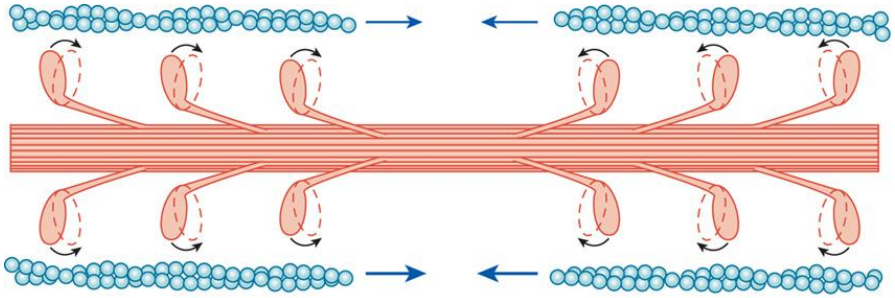
- **Power stroke**
- Stroking (노 젓기) motion pulling the thin filament toward the center of sarcomere
- After regulatory proteins pulled away by Ca^{2+}, the cross bridges from thick filament bind with actin in the thin filament

- **Myosin**
- Motor protein, walking along thin filament to pull it inward
- Only one head of two myosin heads attaches to actin by elevating its head -> myosin head tilts 45 degrees inward (stroking motion) -> detachment and return to original angle -> repeated cross-bridge activity complete the shortening

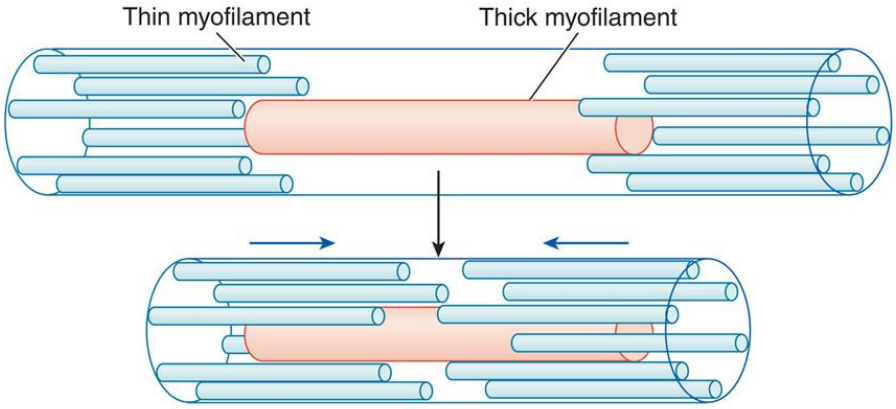


Power stroke (at the thick filament level)

- Not all myosin molecules move synchronously
- Some hold while some detach to enable smooth pulling
- Muscle excitation switch on this cross-bridging cycling by
- **Excitation-contraction coupling**
  (흥분-수축 연결)



(b) All cross-bridge stroking directed toward center of thick filament

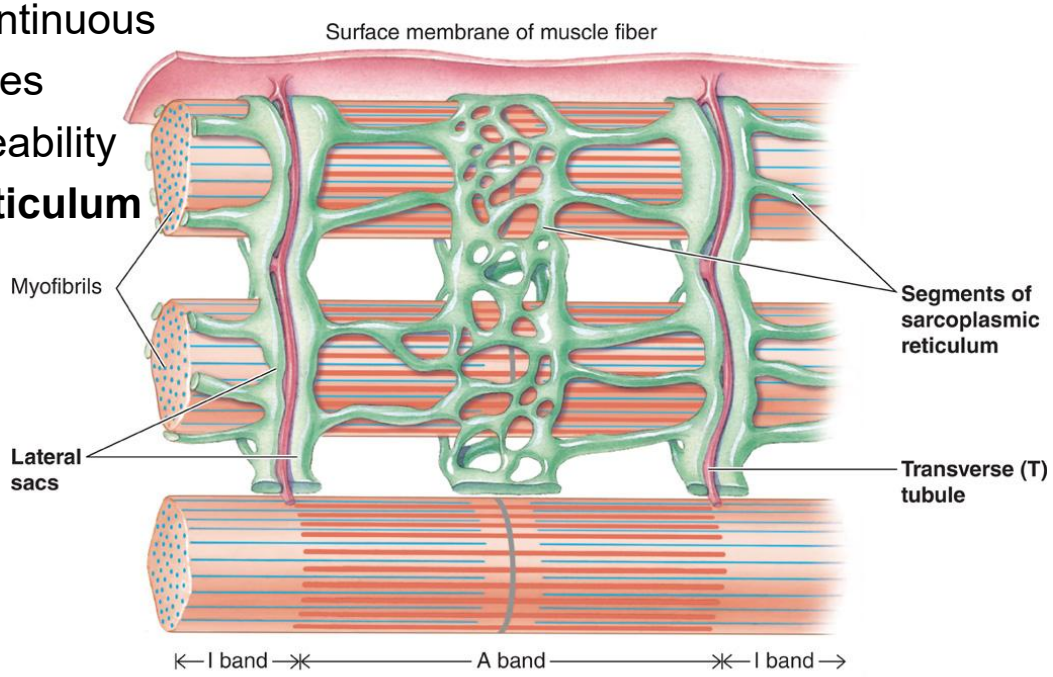


(c) Simultaneous pulling inward of all six thin filaments surrounding a thick filament

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Calcium is the link between excitation and contraction

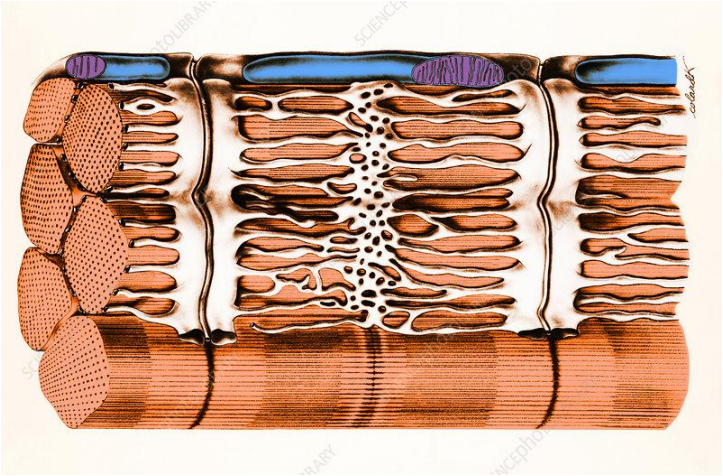
- **Acetylcholine** is released at the neuromuscular junction
- As a result, **action potential** is generated, conducting over the entire surface of the muscle fiber membrane (= sarcolemma)
- **Traverse tubules (T tubule)**
  - At the junction of an A band and I band, the sarcolemma dips into the muscle fiber to form a transverse tubule (T tubule)
  - T tubule plasma membrane is continuous with sarcolemma -> **AP** propagates
  - AP in the T tubule leads to permeability changes in the **sarcoplasmic reticulum**

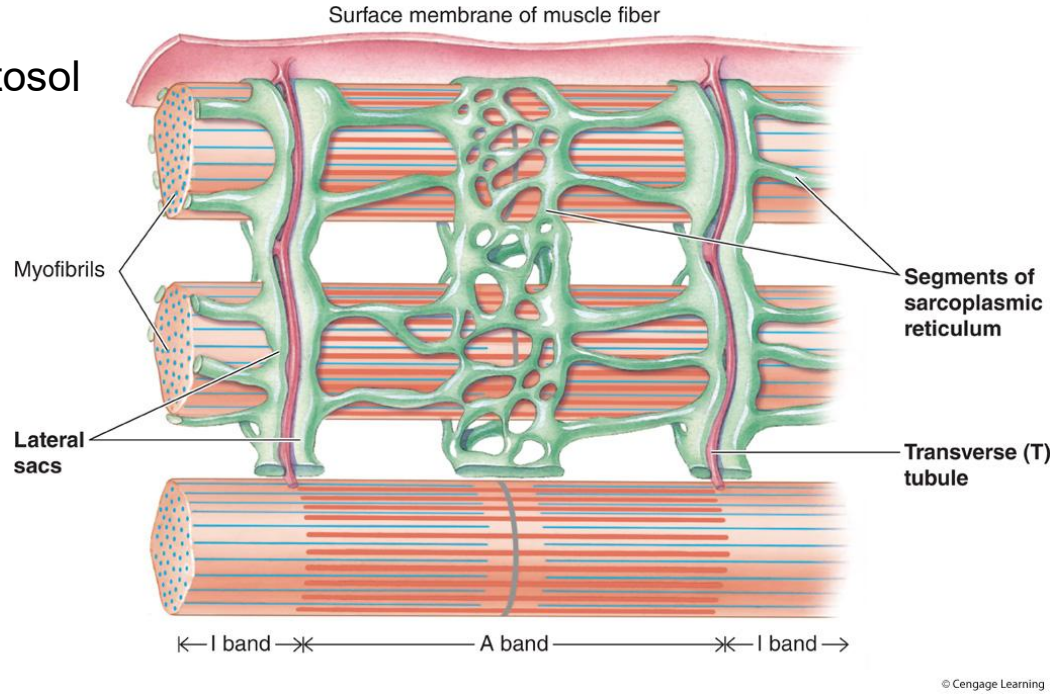


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Sarcoplasmic reticulum releases calcium

- **Sarcoplasmic reticulum** is a modified smooth endoplasmic reticulum consists of fine that forms a mesh sleeve-like network of tubular membranes surrounding each myofibril
- Encircles myofibril around each A band and each I band
- **Lateral sacs**
  - Store Ca^{2+}
  - Spread of AP down a T tubule triggers Ca^{2+} release into the cytosol

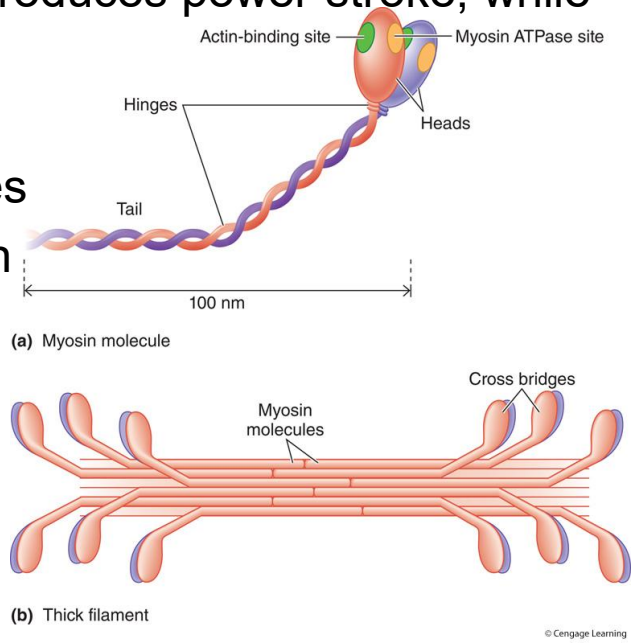




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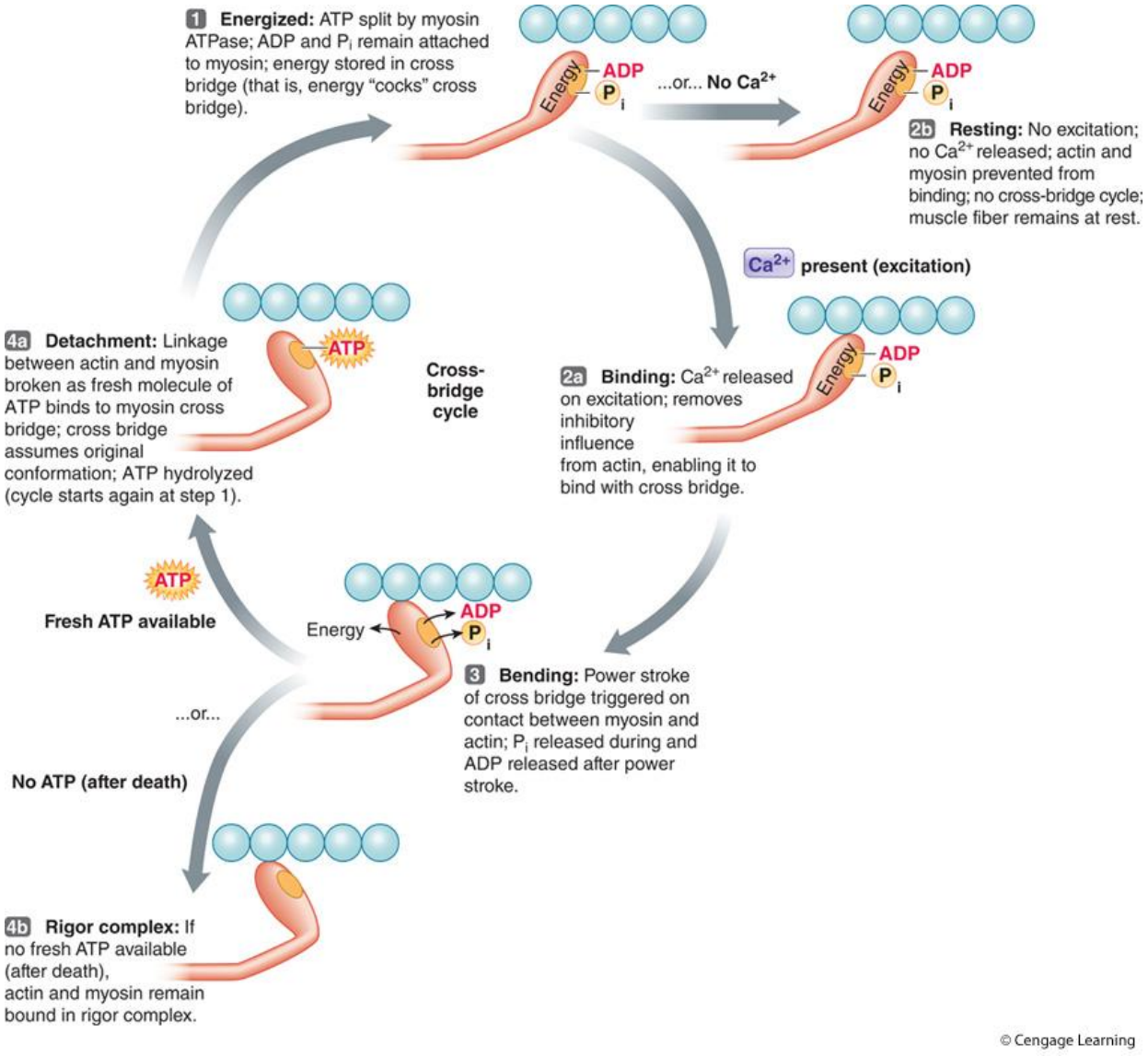
ATP-powered cross-bridge cycling

- A myosin contains actin-binding and **ATPase site**
- Enzymatic site that bind ATP and split into ADP and inorganic phosphate (Pi), generating energy in the process
- ATP breakdown occurs **before** the bridge is linked to the actin -> energy is stored in myosin, in high energy form of myosin
- Muscle fiber excitation (action potential) releases Ca^{2+}, pulling tropomyosin and troponin out of actin filament so that energized myosin can bind to actin
- The contact causes the cross-bridge bending that produces power stroke, while Pi is released during and ADP is released after completion of power stroke
- Actin and myosin remain linked until new ATP comes
- Attachment of new ATP reduces the affinity between actin and myosin, allowing their detachment
- Myosin returns to unbent form
- ATP is split by myosin and cycle repeats



Cross-bridge cycle

rigor mortis
(사후 경직)



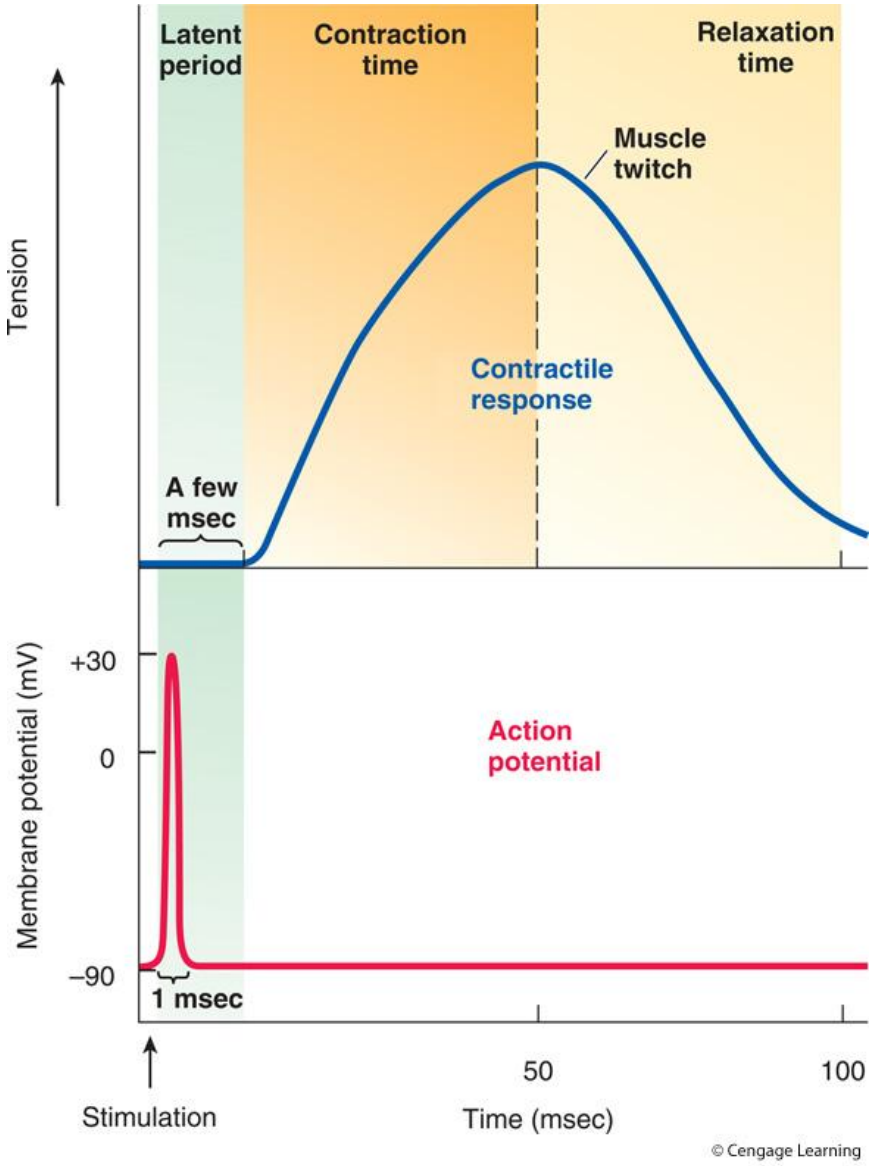
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Relaxation and latent period

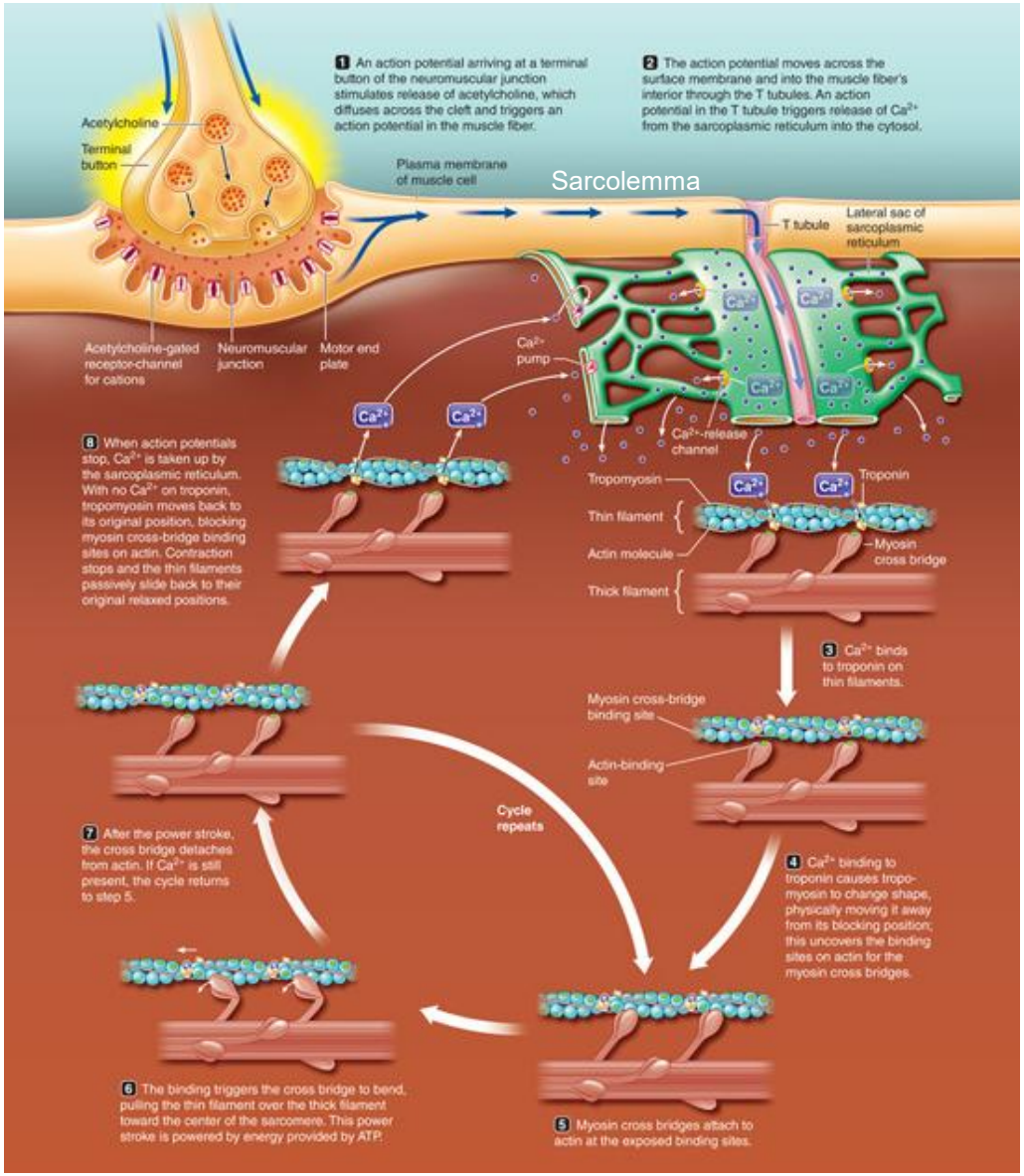
- Relaxation
  - When local electrical activity is stopped, Ca^{2+} is returned to the lateral sacs of SR and contraction process is turned off
  - Sarcoplasmic/endoplasmic reticulum Ca^{2+}-ATPase **(SERCA) pump**: transports Ca^{2+} to the lateral sacs
  - When acetylcholinerase at the NMJ removes Ach, action potential no longer persists and SERCA pump returns Ca^{2+} back to the SR
  - Myosin can no longer bind to actin

- Contractile activity lasts for a while until relaxation occurs
  - A single action potential lasts only 1 to 2 ms
  - The contractile latent period occurs because excitation-contraction coupling steps (Ca^{2+} release and cross-bridge activation) take time after the action potential
  - Contraction time varies from 15 to 50 ms: time needed for full tension generation
  - Relaxation time also takes 15 to 50 ms: Ca^{2+} must be fully pumped back into the SR, leading to dissociation of cross-bridges
  - Important for body's ability to produce muscle contraction of variable strength

Action potential and muscle contraction time



Excitation-contraction coupling



Lateral sac: part of sarcoplasmic reticulum, store Ca^{2+} when muscle is at rest

Cross-bridge formation continues while Ca^{2+} presents (ATP is required per cycle)

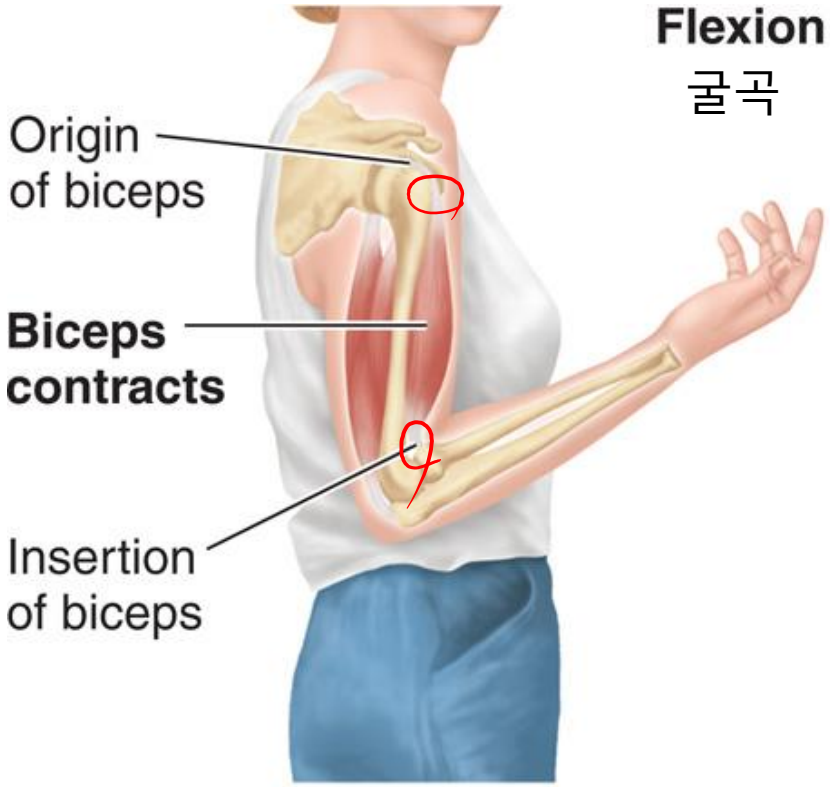
8.3 Skeletal Muscle Mechanics

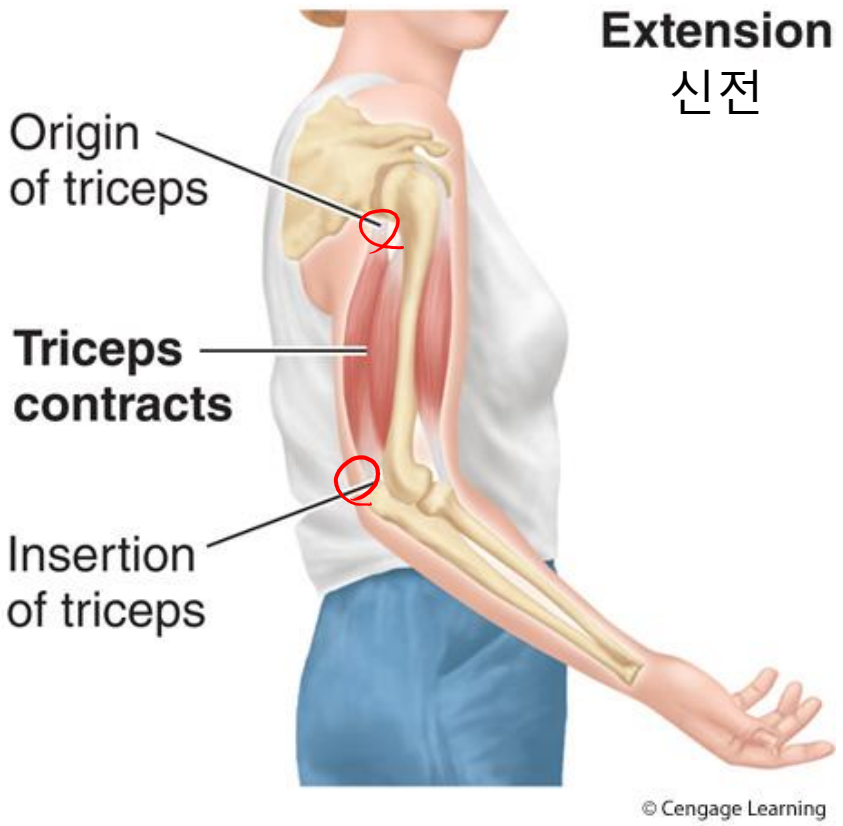
Skeletal muscle mechanics

- Whole muscles are groups of muscle fibers bundled together and attached to bones
- Connective tissues divide the muscle into the bundles (fascicles)
- Tendons attach the muscle to bones

- Series-elastic component of muscle (직렬 탄성 요소)
- Muscle **tension** is produced in the sarcomeres (contractile unit of muscle) and transmitted to bone via tendon before the bone is moved
- The series-elastic component (SEC) in muscle mechanics refers to tendons acting in series with muscle fibers, behaving as **springs** that store and release elastic energy to enhance power and efficiency during movement
- Tendons absorb shock, reduce metabolic cost during walking/running, and protect muscles during rapid lengthening

Flexion and extension of elbow joint

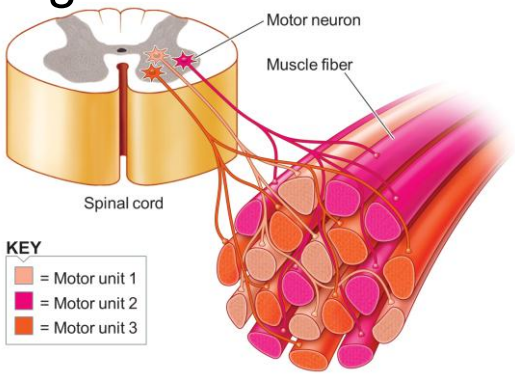




Origin: the end of muscle attached to the more stationary part of the skeleton
Insertion: the end attached to the skeletal part that moves

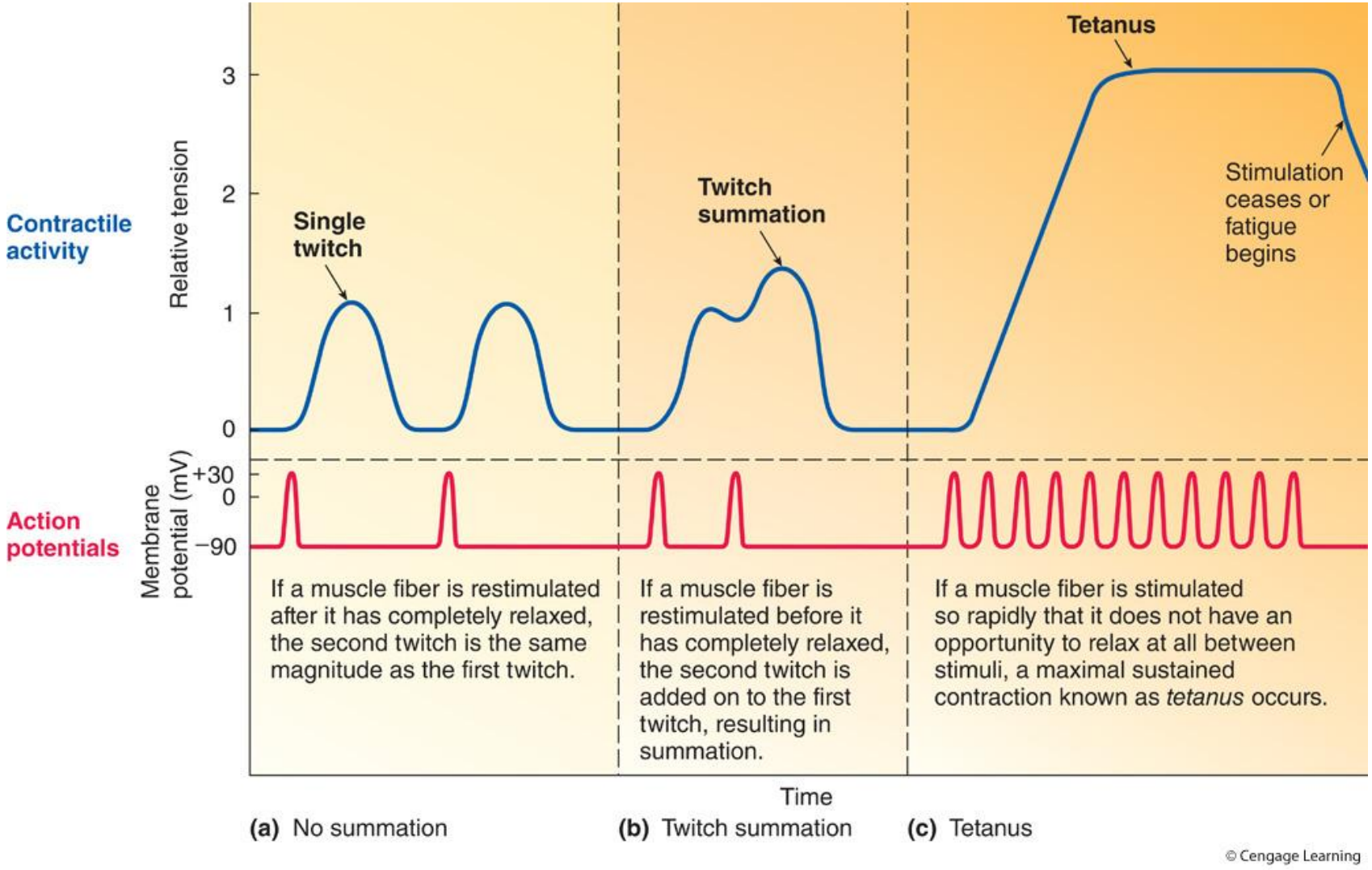
Twitch (경련)

- **Twitch**: a brief and weak contraction generated by single action potential in a muscle fiber
- Gradation of whole muscle tension is generated by
  - The number of muscle fibers contracting within a muscle
  - The tension developed by each contracting fiber



- The number of fibers contracting within a muscle is depends on how many **motor units** are recruited
  - Motor unit: one motor neuron from spinal cord + all the muscle fibers it innervates
- Twitch summation: the increase in tension after repetitive stimulation of a muscle fiber
  - Successive action potentials arriving before complete relaxation produce summation of muscle tension
  - If a muscle fiber is restimulated before complete relaxation, second stimuli generates greater contraction (via sustained elevation of cytosolic Ca^{2+})
- Tetanus (경직): sustained contraction of maximal strength
  - If a muscle fiber is stimulated so rapidly without time for relaxation, a maximal sustained contraction occurs

Twitch (경련)

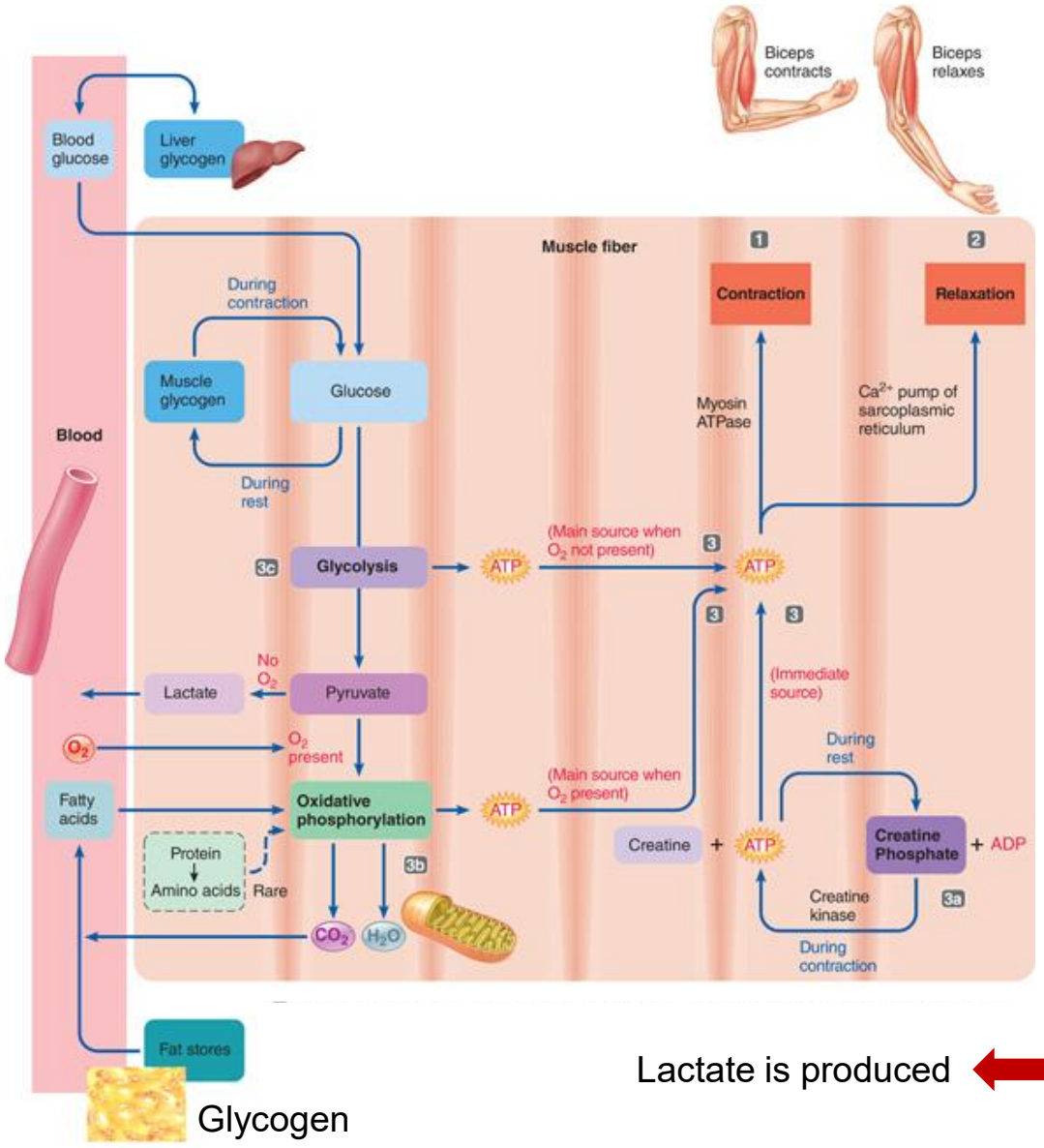

Contractile activity
Relative tension
3
2
1
0
Single twitch
Twitch summation
Tetanus
Stimulation ceases or fatigue begins
Action potentials
Membrane potential (mV)
+30
0
−90
If a muscle fiber is restimulated after it has completely relaxed, the second twitch is the same magnitude as the first twitch.
If a muscle fiber is restimulated before it has completely relaxed, the second twitch is added on to the first twitch, resulting in summation.
If a muscle fiber is stimulated so rapidly that it does not have an opportunity to relax at all between stimuli, a maximal sustained contraction known as tetanus occurs.
Time
(a) No summation
(b) Twitch summation
(c) Tetanus
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8.4 Skeletal Muscle Metabolism and Fiber Types

Muscle fibers can generate ATP by alternative route

- ATP is required by muscles in four steps
  - ATP splitting by myosin ATPase site provides the **energy for power stroke** of cross-bridge
  - Binding of new ATP to myosin breaks cross-bridge
  - Active transport of Ca^{2+} back into lateral sacs during relaxation
  - Activity of Na^{+}-K^{+} pump during action potential production
- ATP is the only direct energy source for these activities
- Limited storage of ATP is available in muscle, only enough to supply power for first few seconds of exercise
- Three pathways supply additional ATP needed for muscle contraction
  - Creatine phosphate
  - Oxidative phosphorylation
  - Glycolysis

Muscle fibers can generate ATP by alternative route



1 During **muscle contraction,** ATP is split by myosin ATPase to power cross-bridge stroking. Also, a fresh ATP must bind to myosin to let the cross bridge detach from actin at the end of a power stroke before another cycle can begin.

2 During **relaxation,** ATP is needed to run the Ca^{2+} pump that transports Ca^{2+} back into the lateral sacs of the sarcoplasmic reticulum. ATP is also used by the Na^+–K^+ pump to return Na^+ and K^+ moved during contraction-inducing action potentials.

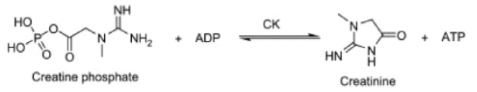
3 The **metabolic pathways that supply the ATP** needed to accomplish contraction and relaxation are

3a transfer of a high-energy phosphate from **creatine phosphate** to ADP (immediate source);

3b **oxidative phosphorylation** (the main source when O_2 is present), fueled by glucose derived from muscle glycogen stores or by glucose and fatty acids delivered by the blood; and

3c **glycolysis** (the main source when O_2 is not present). Pyruvate, the end product of glycolysis, is converted to lactate when lack of O_2 prevents the pyruvate from being further processed by the oxidative phosphorylation pathway.

1. Creatine phosphate
 - first energy storehouse similar to ATP
 - ready to donate Pi to ADP
 - creatine kinase



2. Oxidative phosphorylation
 - multi-step process taking more time
 - use fatty acid or glucose
 - in muscle mitochondria if oxygen is present
 - myoglobin transfers and stores O_2 from blood

3. Glycolysis (anaerobic)
 - main source when O_2 is not present
 - rapid ATP production than OXPHOS
 - glucose -> two pyruvates + 2 ATP
 - pyruvate is used for OXPHOS

Lactate is produced

Three types of skeletal muscle fibers

- Two main differences among three fiber types
  - speed of contraction (slow or fast)
  - enzymatic machinery primarily used (OXPHOS or glycolysis)

- Slow-oxidative fiber (type I)
- Fast-oxidative fiber (type IIa)
- Fast-glycolytic fiber (type IIx)

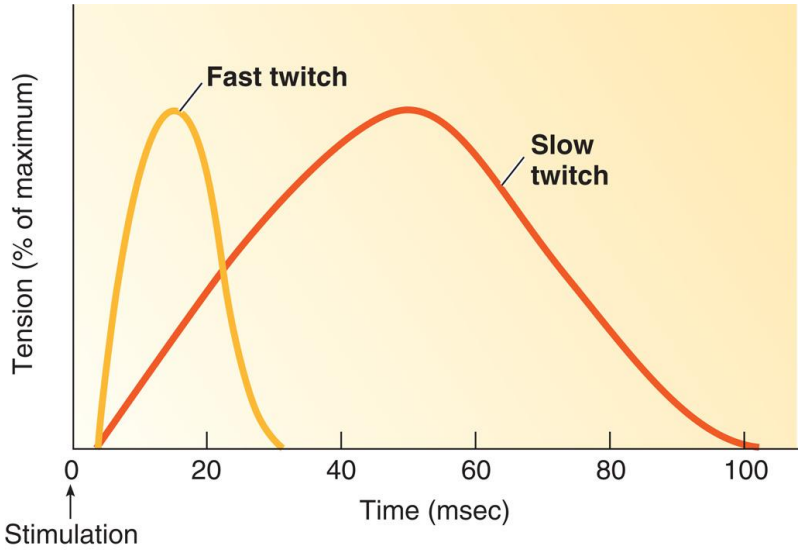
- Fast fibers have higher myosin ATPase activity than slow fibers -> Fast ATP split -> Fast cross-bridge cycling -> Fast twitch
- Primarily rely on OXPHOS or glycolysis for ATP synthesis
  - Oxidative fibers (red fiber) contain more mitochondrial to yield more ATP -> enough ATP due to high capacity for aerobic ATP production, less lactate accumulation (resistant to fatigue)
  - Glycolytic fibers store more glycogen and glycolytic enzymes

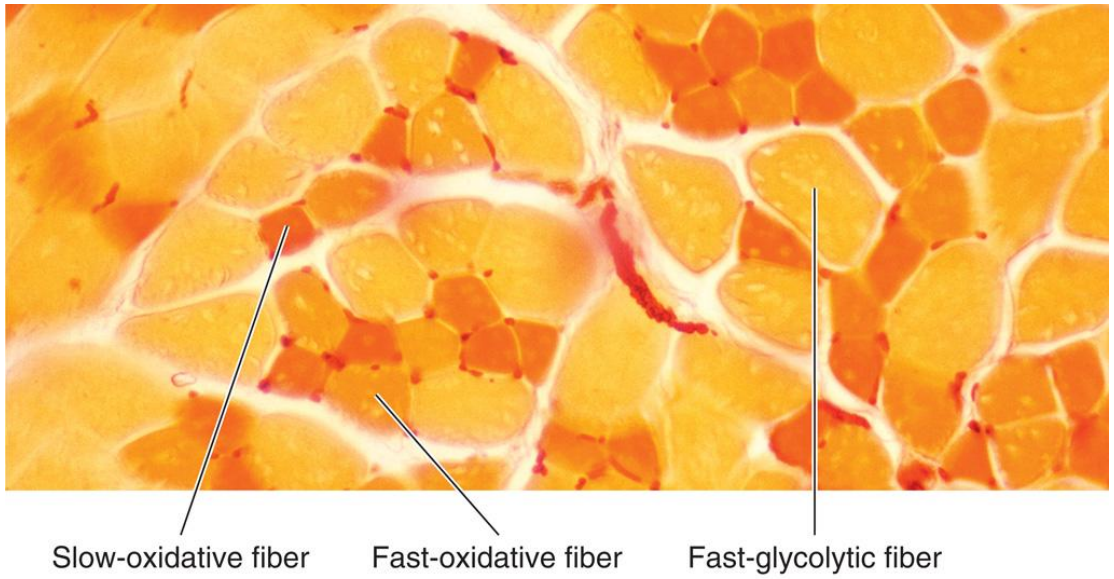
Three types of skeletal muscle fibers

TABLE 8-1 Characteristics of Skeletal Muscle Fibers

Characteristic	Slow-Oxidative (Type I) Fiber	Fast-Oxidative (Type IIa) Fiber	Fast-Glycolytic (Type IIx) Fiber
Myosin–ATPase activity	Low	High	High
Speed of contraction	Slow	Fast	Fast
Resistance to fatigue	High	Intermediate	Low
Oxidative phosphorylation capacity	High	High	Low
Enzymes for anaerobic glycolysis	Low	Intermediate	High
Mitochondria	Many	Many	Few
Capillaries	Many	Many	Few
Myoglobin content	High	High	Low
Color of fiber	Red	Red	White
Glycogen content	Low	Intermediate	High

Three types of skeletal muscle fibers





8.5 Control of Motor Movement

Control of motor movement

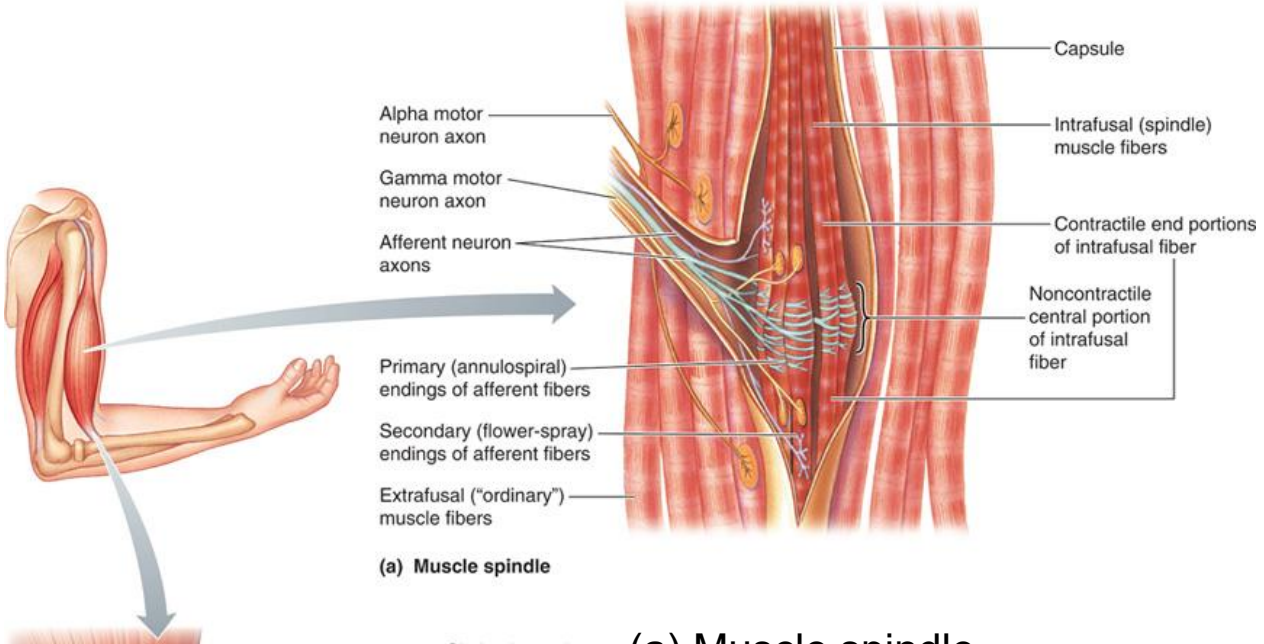
- Motor activity can be classified as reflex, voluntary, or rhythmic
  - Reflex: automatic response by skeletal muscle contraction without conscious effort, integrated in the spinal cord (spinal) or brain stem (cranial)
  - Voluntary movement: complex info. integrated by cerebral cortex
  - Rhythmic activity: movements repeated in general pattern like walking and chewing, starts consciously and maintains without conscious effort
- Multiple neural inputs influence motor unit output
  - Input from afferent neurons: spinal reflexes
  - Input from primary motor cortex: pyramidal cells within primary motor cortex (Chap. 5) make orders
  - Input from **brain stem**: receives input from motor cortex, cerebellum, basal nuclei, and afferent neurons, serves as final link between higher motor centers and skeletal muscle contraction

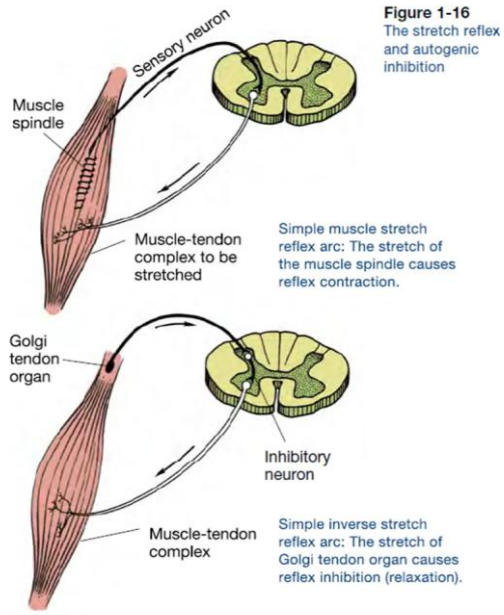
Muscle receptors

- **Muscle receptors** provide afferent information needed to control skeletal muscle activity
- Before ordering muscle activity, CNS must know the starting position and location of the body -> **proprioceptor**
- Proprioceptor: sensory nerve ending in muscles, tendons, and joints detecting body movement, position, and tension
- Provide information about location of body parts and body movement -> continuous information about muscle length and tension

- Muscle spindles: monitor muscle length
- Golgi tendon organs: detect changes in tension

Three types of skeletal muscle fibers





(a) Muscle spindle
- Intrafusal fibers (방추 속근) with primary and secondar afferent endings
- Detect **muscle length** and stretch to regulate reflex

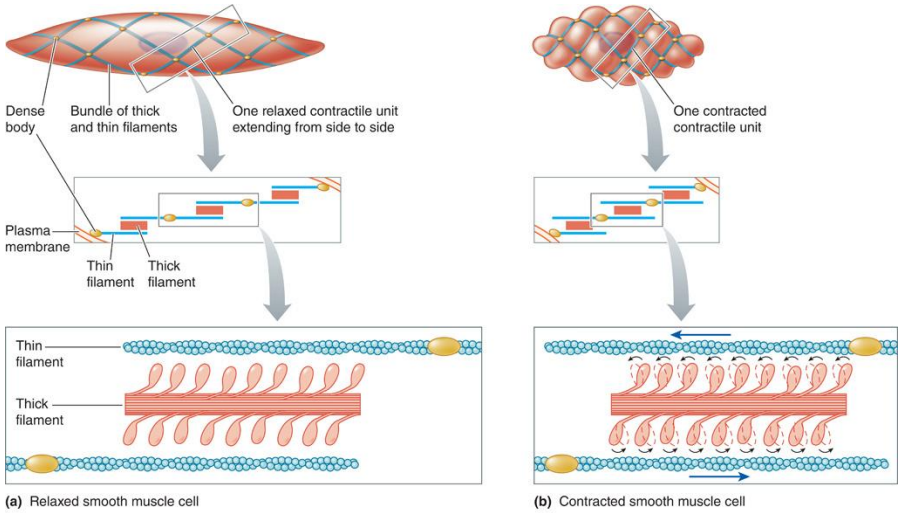
(b) Golgi tendon organ
- Collagen fibers with afferent endings located in a tendon
- Detect **muscle tension** and trigger reflex relaxation

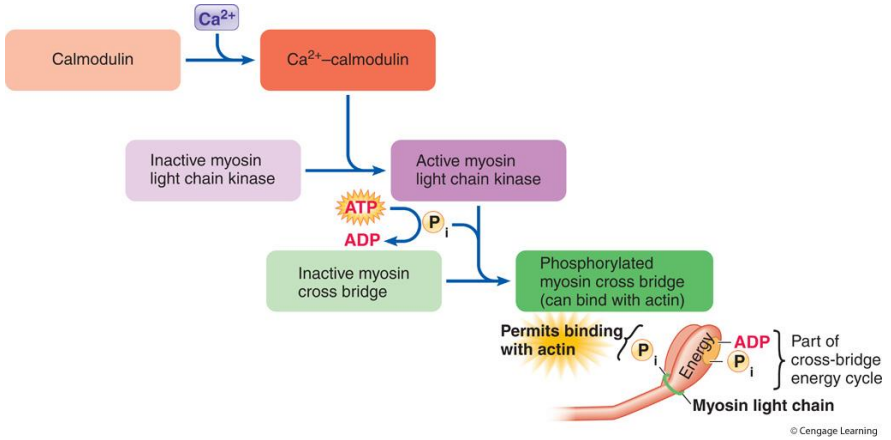
8.6 Smooth and Cardiac Muscle

Smooth muscle

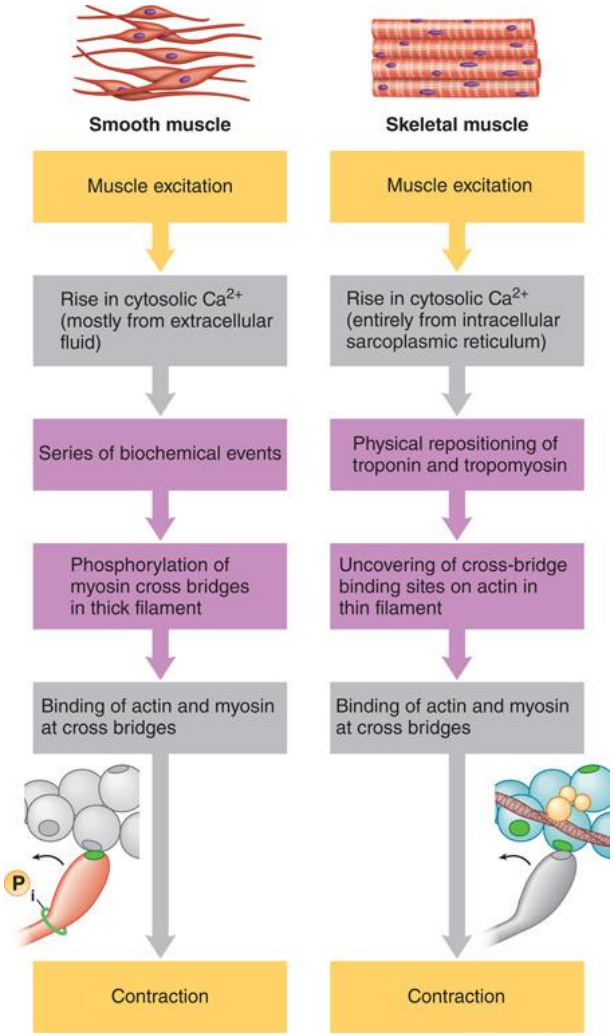
- Three muscle types all have contractile apparatus made up of thin actin filaments and myosin filaments, they all use ATP for cross-bridge cycling
- Smooth muscle cells
  - Are small and unstriated (no stripes)
  - Are turned on by Ca^{2+}-dependent phosphorylation of myosin, not by Ca^{2+} binding to troponin
  - Slow but economic (single contraction lasts 3 s -> less ATP use)

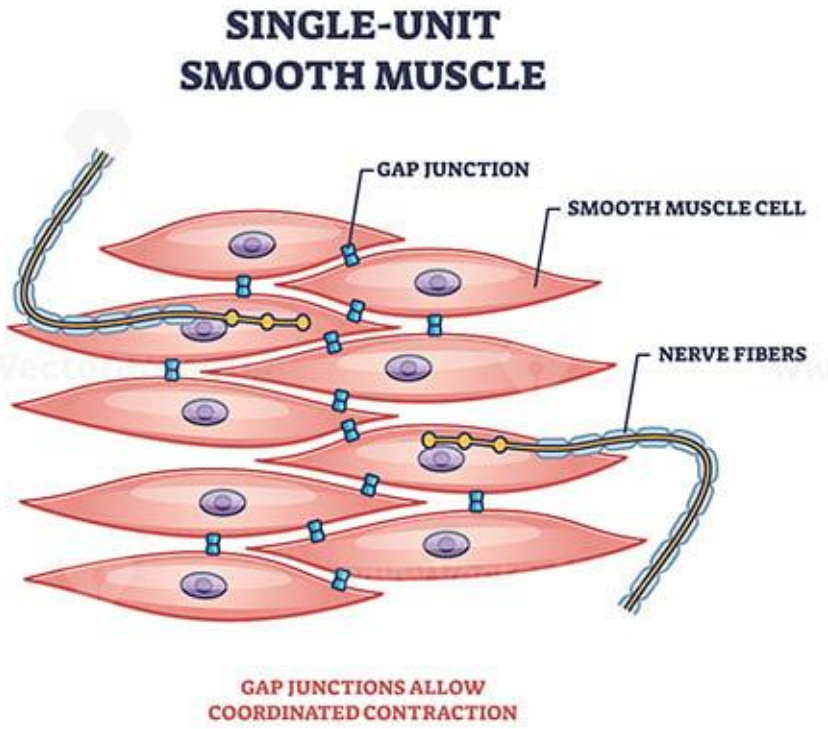
Found in the walls of hollow organs, blood vessels, and tubular structures (digestive, respiratory, urinary, reproductive tracts, eyes)





Smooth muscle





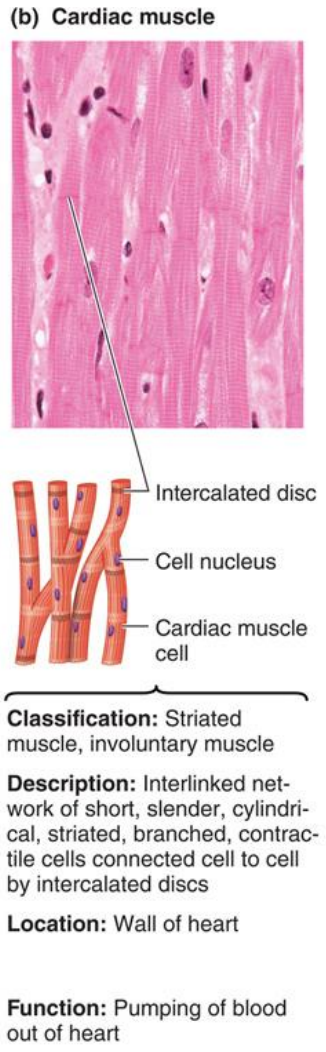
Single-unit smooth muscle
- Collection of smooth muscle cells connected by gap junctions
- AP can propagate, coordinating muscle contraction as a unit
- Self-excitable, without neuron

Cardiac muscle

- Found only in the heart
- Innervated by the autonomic nervous system
- Cardiac muscle mixes features of skeletal and smooth

- Highly organized into banding pattern (striated), slender, short fibers
- Contain troponin and tropomyosin
- Has a lot of mitochondria and myoglobin, T tubules, and sarcoplasmic reticulum, like skeletal muscle

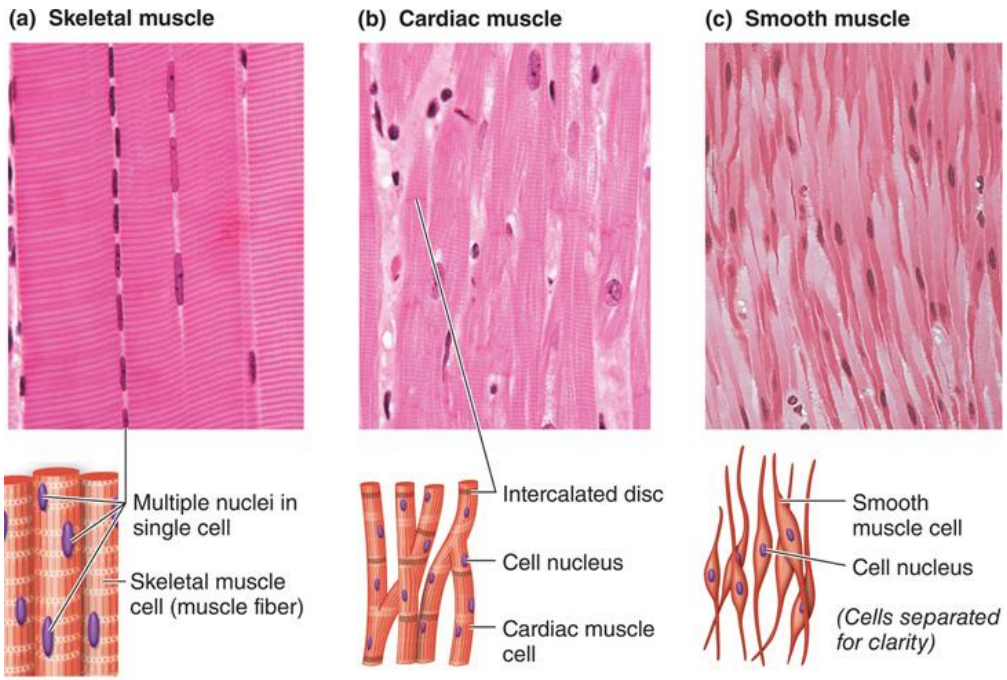
- Short and slender, can initiate AP without external influence, like smooth muscle
- Contains gap junctions found in intercalated disc joining cells together
- Long refractory period prevents tetanus



Three types of muscles

	Skeletal	Cardiac	Smooth
Mechanism of contraction	Sliding filament mechanism (myosin and actin filaments)		
Nucleus	Many	Single	
Neural innervation	Somatic nervous	Autonomic nervous system	
Level of control	Voluntary + subconscious control	Under involuntary control	
Initiation of contraction	Neurogenic	Myogenic (mostly)	
Gap junction	No	Yes	
Contraction and relaxation	Rapid	Slow, self-excited, rhythmic	Very slow, self-excited, rhythmic
Ca^{2+} role	Troponin & tropomyosin		Calmodulin – myosin light chain kinase
T tubules	Yes		No
Sarcoplasmic reticulum	Well developed		Poorly developed
Striation (thick and thin)	Yes		No
Filament arrangement	Highly organized in sarcomeres		Dispersed, no sarcomeres; anchored to dense bodies
Location	Skeletal muscles	Heart walls	Hollow visceral walls, blood vessels

Characteristics of three types of muscle



Classification: Striated muscle, voluntary muscle

Description: Bundles of long, thick, cylindrical, striated, contractile, multinucleate cells that extend the length of the muscle

Typical location: Attached to bones of skeleton

Function: Movement of body in relation to external environment

Classification: Striated muscle, involuntary muscle

Description: Interlinked network of short, slender, cylindrical, striated, branched, contractile cells connected cell to cell by intercalated discs

Location: Wall of heart

Function: Pumping of blood out of heart

Classification: Unstriated muscle, involuntary muscle

Description: Loose network of short, slender, spindle-shaped, unstriated, contractile cells that are arranged in sheets

Typical location: Walls of hollow organs and tubes, such as stomach and blood vessels

Function: Movement of contents within hollow organs

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